

Lab Assignment 1: Flowchart and Java Programming

1. Check Positive Number:

1.1 Flowchart: Num = 2

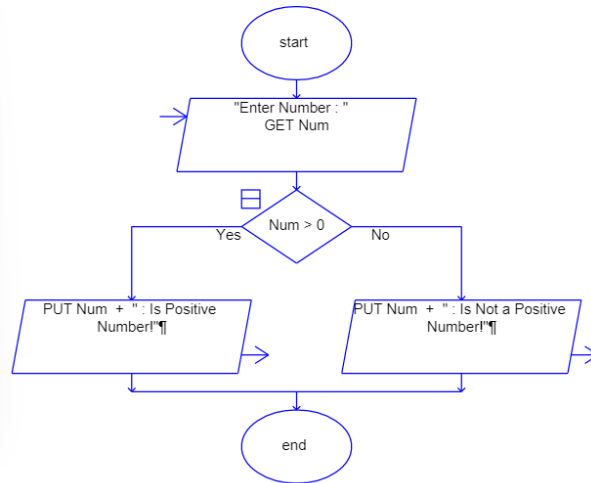
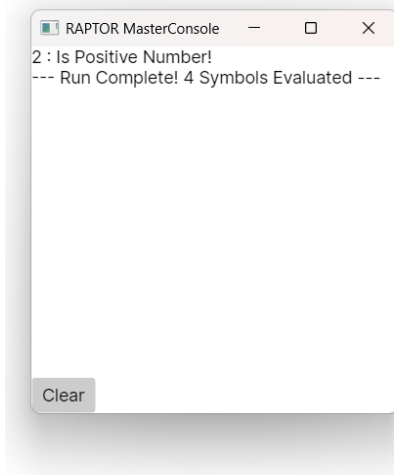


Fig.1.1 Check Positive Number Flowchart

1.2 Simple Java Program

1.2.1 Program :

```
class PositiveNum
{
    public static void main(String[] args) {
        int num = 10;

        if (num > 0)
            System.out.println(num + " is a positive number.");
        else
            System.out.println(num + " is not a positive number.");
    }
}
```

1.2.2 Code Output :

```
C:\Windows\System32\cmd.e  ×  +  ∨

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac PositiveNum.java

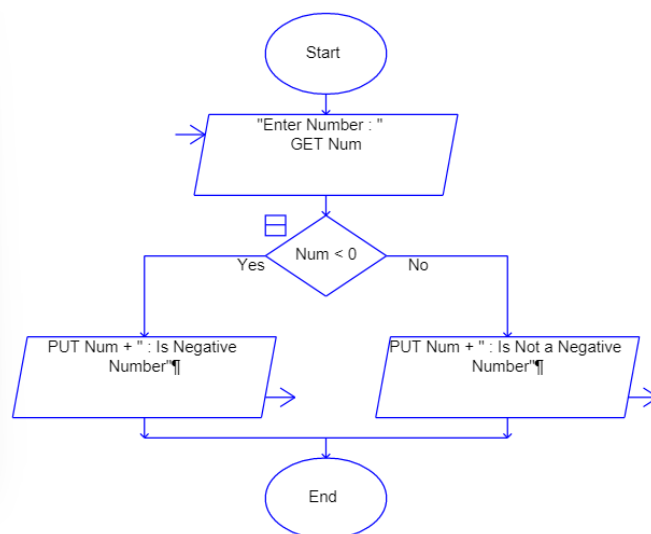
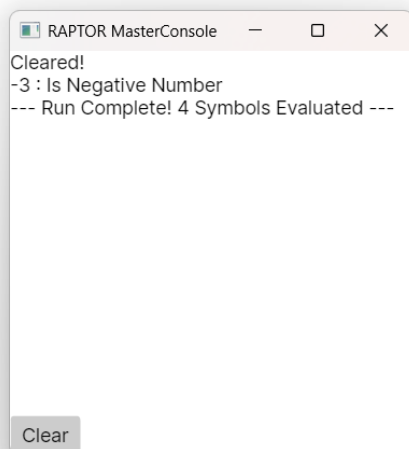
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java PositiveNum
10 is a positive number.

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

Fig.1.2.2 Positive Number Output

2. Check Negative Number

2.1 Flowchart of Negative Number – Num = -3



2.1 Flowchart of Negative Number

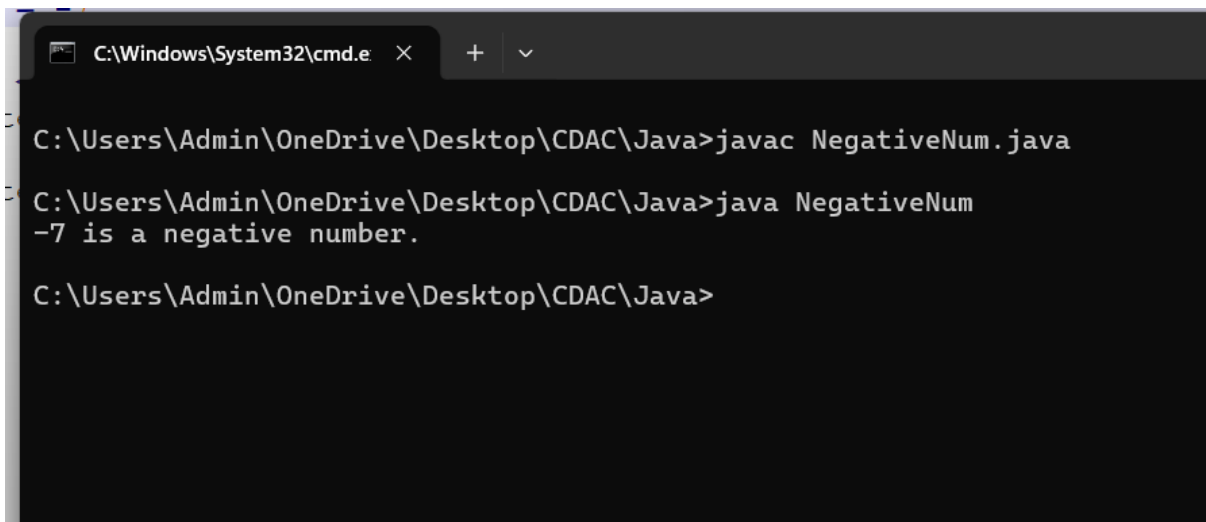
2.2 Simple Java Program of Negative number

2.2.1 Code

```
public class NegativeNum
{
    public static void main(String[] args) {
        int num = -7;

        if (num < 0)
            System.out.println(num + " is a negative number.");
        else
            System.out.println(num + " is not a negative number.");
    }
}
```

2.2.2 Output of Negative number



The screenshot shows a Windows command prompt window with the title bar 'C:\Windows\System32\cmd.e'. The command history is as follows:

```
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac NegativeNum.java
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java NegativeNum
-7 is a negative number.
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

Fig.2.2.2 Output of Negative number

3. Check Odd or Even Number

3.1 Flowchart – Num=6

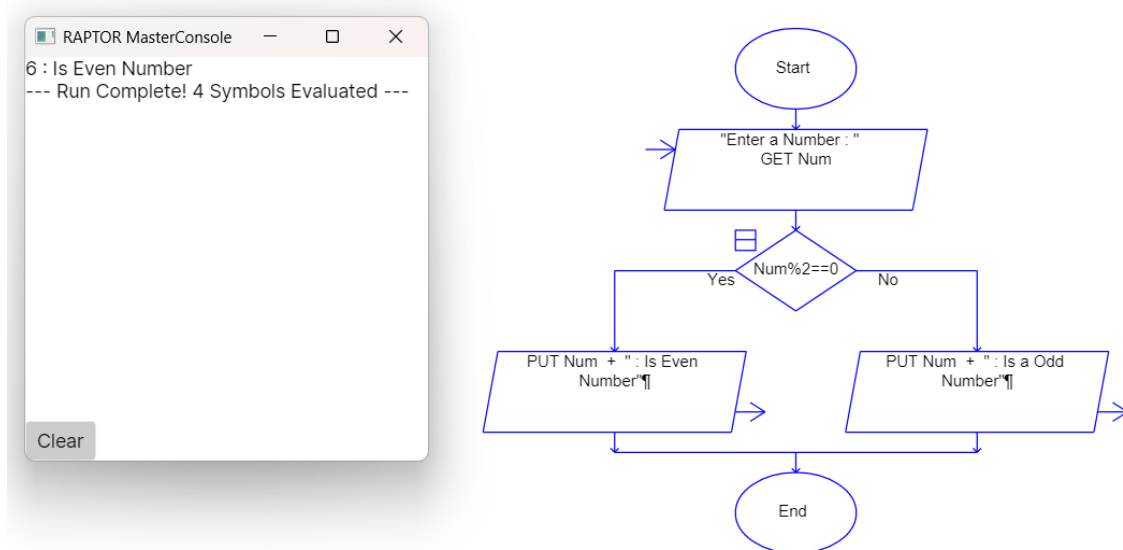


Fig.3.1 Flowchart of Odd or Even

3.2 Simple Java code

3.2.1 Code Even or odd

```
public class EvenOdd {
    public static void main(String[] args) {
        int num = 6;

        if (num % 2 == 0)
            System.out.println(num + " is even.");
        else
            System.out.println(num + " is odd.");
    }
}
```

3.2.2 Output of Even or odd

```
C:\Windows\System32\cmd.e x + v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac EvenOdd.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java EvenOdd
6 is even.

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

Fig.3.2.2 Output of Even or odd

4. Display Good Morning Message Based on Time

4.1 Flowchart

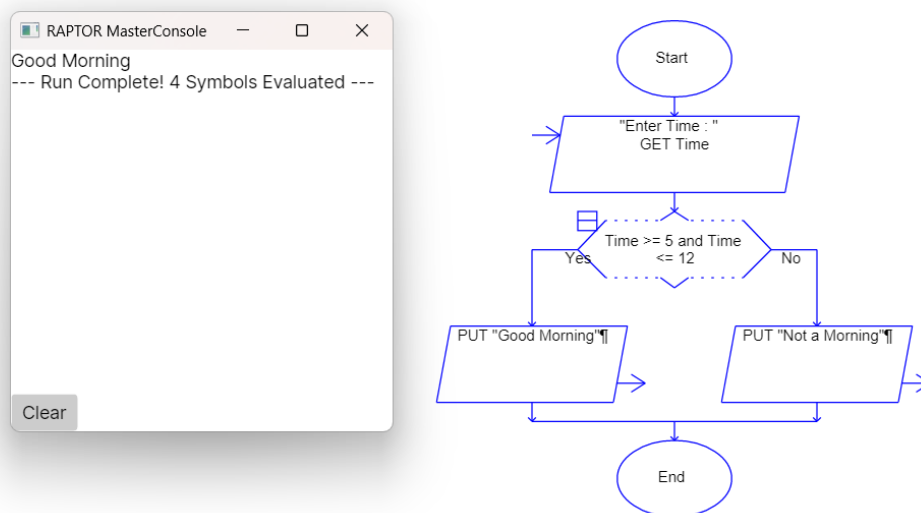


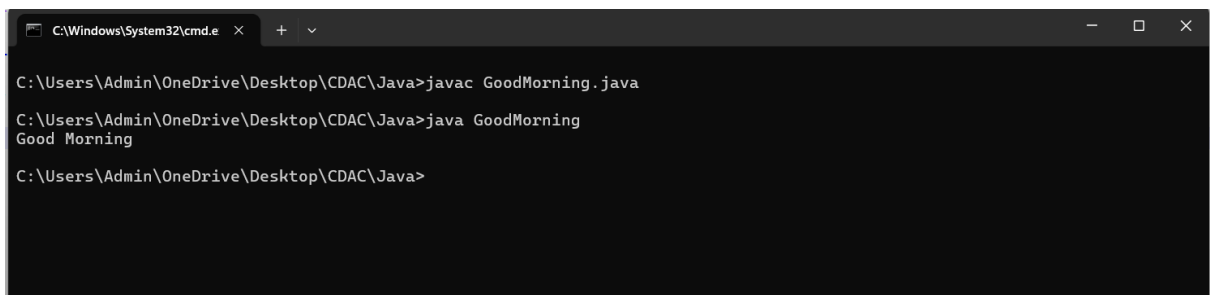
Fig.4.1 Flowchart of Good Morning based on time

4.2 Simple Java code of Good Morning based on time

4.2.1 Code

```
public class GoodMorning {  
    public static void main(String[] args) {  
  
        int time = 9; // predefined time  
  
        if (time >= 5 && time <= 12) {  
            System.out.println("Good Morning");  
        }  
    }  
}
```

4.2.2 Output of the code



```
C:\Windows\System32\cmd.e  X + v  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac  GoodMorning.java  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java GoodMorning  
Good Morning  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

Fig. 4.2.2 Output of the code

5. Print Area of a Square

5.1 Flowchart

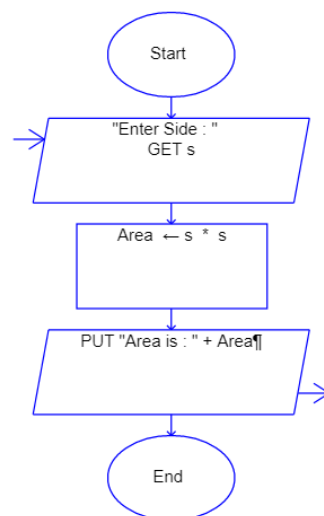
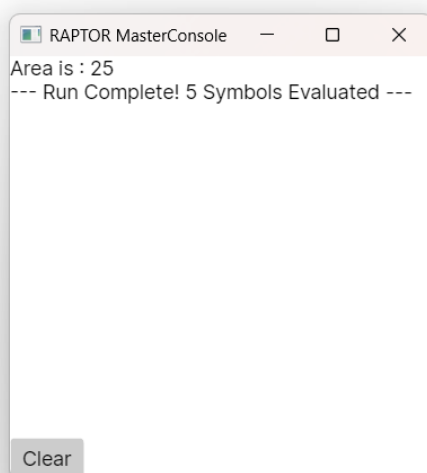


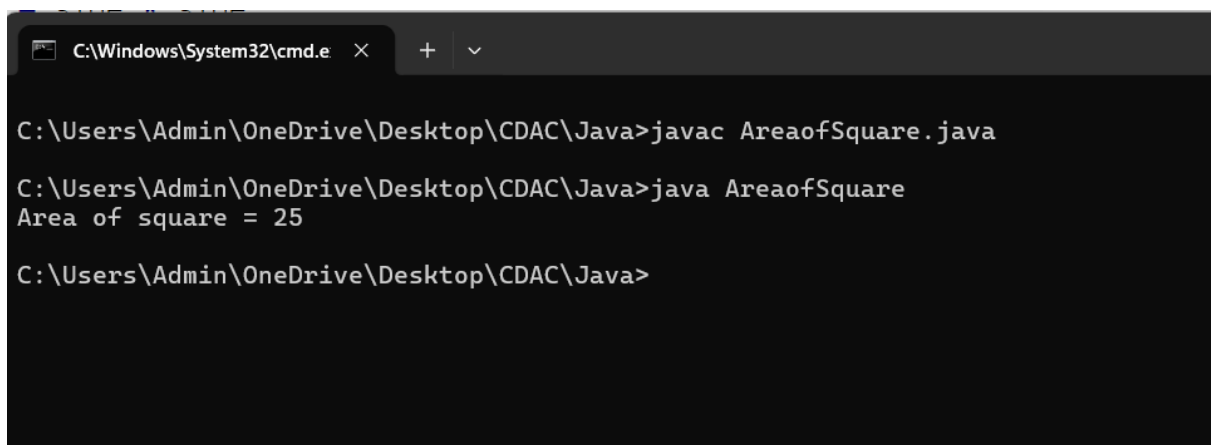
Fig.5.1 Flowchart of Area of a Square

5.2 Simple program of Area Of a Square

5.2.1. code of Area Of a Square

```
public class AreaofSquare {  
    public static void main(String[] args) {  
        int side = 5;  
        int area = side * side;  
  
        System.out.println("Area of square = " + area);  
    }  
}
```

5.2.2 Output of Area of a Square

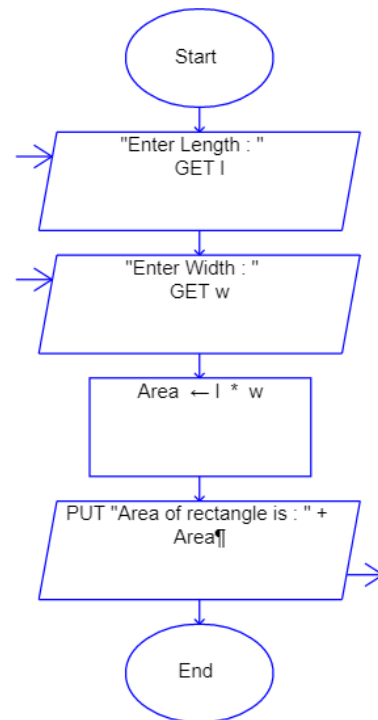
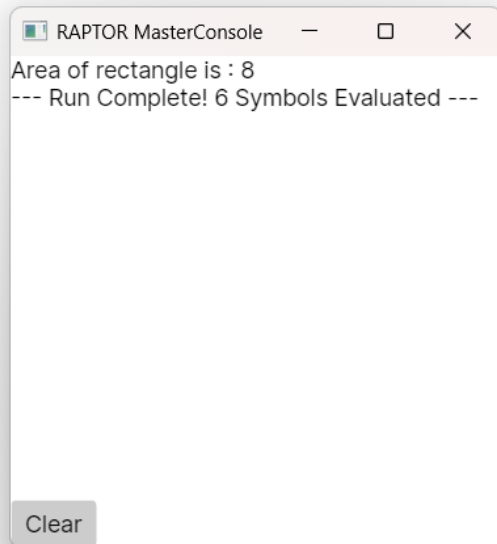


```
C:\Windows\System32\cmd.e  X  +  v  
  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac AreaofSquare.java  
  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java AreaofSquare  
Area of square = 25  
  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

Fig.5.2.2 Output of Area of a Square

6. Print Area of a Rectangle

6.1 Flowchart of Area of a Rectangle



6.1 Flowchart of Area of a Rectangle

6.2 Simple java code of Area of a Rectangle

6.2.1 Code :

```
public class AreaofRectangle {  
    public static void main(String[] args) {  
        int length = 4;  
        int width = 2;  
        int area = length * width;  
  
        System.out.println("Area of rectangle = " + area);  
    }  
}
```


6.2.2 Output of Area of a Rectangle

```
C:\Windows\System32\cmd.e x + v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac AreaofRectangle.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java AreaofRectangle
Area of rectangle = 8

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

Fig.6.2.2 code of Area of a Rectangle

7. Find the Largest of Three Numbers

7.1 Flowchart of the Largest of Three Numbers

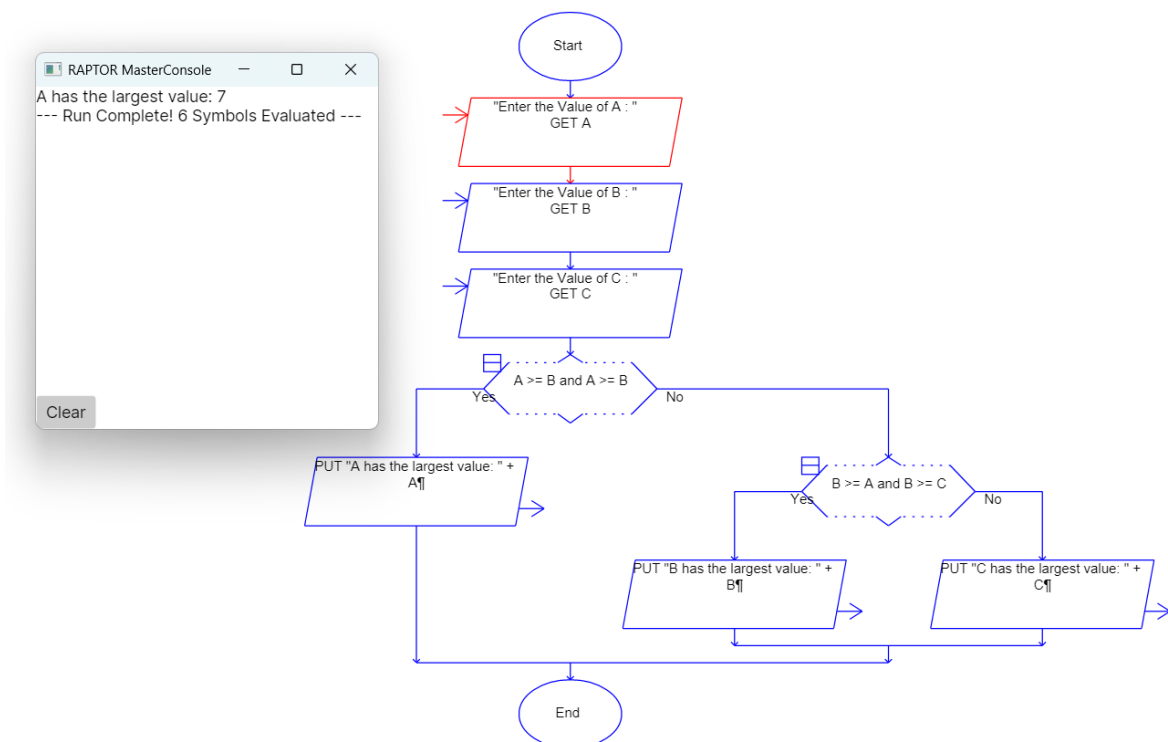


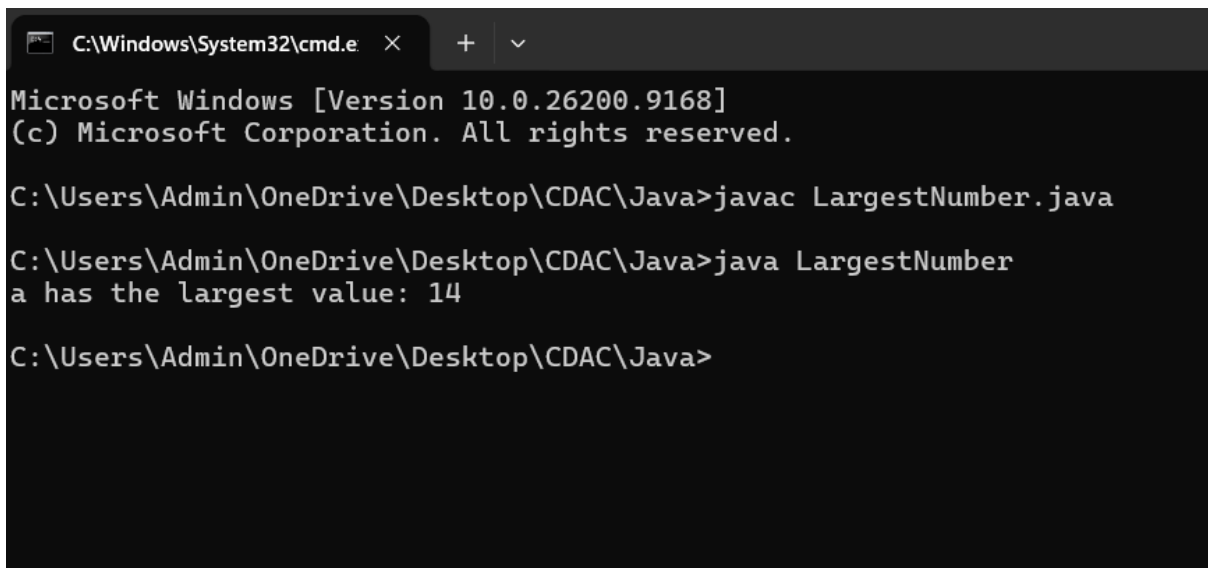
Fig. 7.1 Flowchart of the Largest of Three Numbers

7.2 Simple java Program of the Largest of Three Numbers

7.2.1.Code

```
public class LargestNumber {  
    public static void main(String[] args) {  
        int a = 14;  
        int b = 4;  
        int c = 11;  
  
        if (a > b && a > c)  
            System.out.println("a has the largest value: " + a);  
        else if (b > c && b > a)  
            System.out.println("b has the largest value: " + b);  
        else  
            System.out.println("c has the largest value: " + c);  
    }  
}
```

7.2.2.Output



```
C:\Windows\System32\cmd.e  ×  +  ▾  
Microsoft Windows [Version 10.0.26200.9168]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac LargestNumber.java  
  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java LargestNumber  
a has the largest value: 14  
  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

Fig.7.2.2 Output of the Largest of Three Numbers